Spatial and temporal patterns of Uber trip demand around New York City

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Agenda

01 → 02 → 03 → 04 → 05

01 Motivation & Research Question

02 Data & Approach

03 Key Results

04 Live Demo

05 Q&A

01 Research Question & Motivation

RQ 1

How is **Uber trip demand** distributed across **space** and **time** in the New York and New Jersey states?

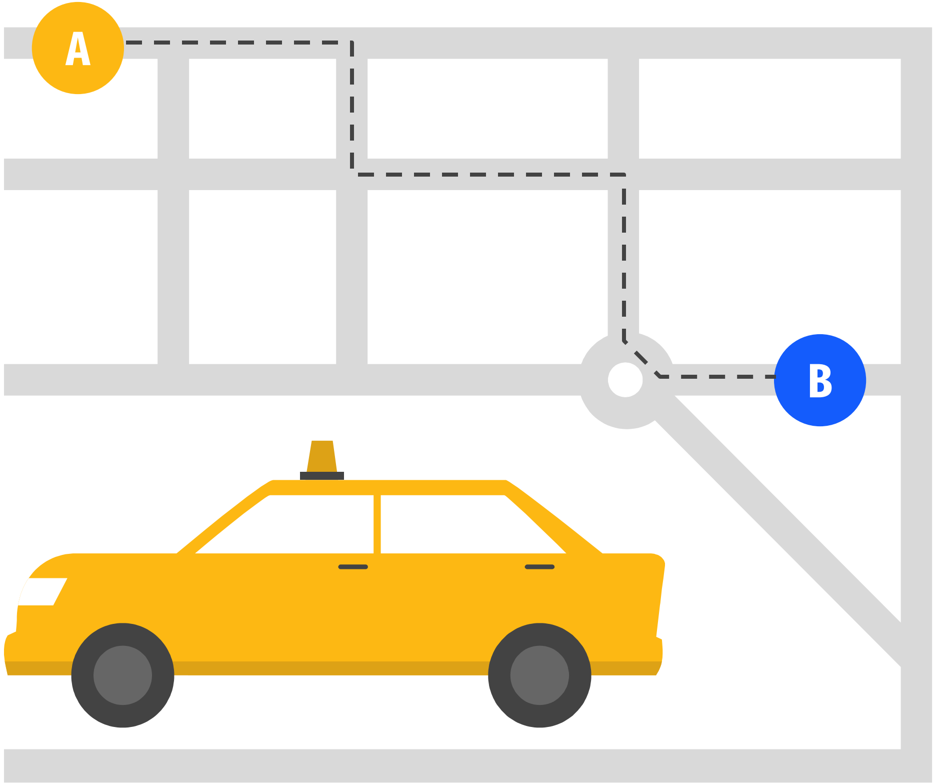
RQ 2

Can **location-based clusters** reveal distinct **daily and hourly demand patterns** that may also be useful for future **forecasting**?

Better understanding of travel behavior in large city

How data-driven methods can support transportation analysis and forecasting

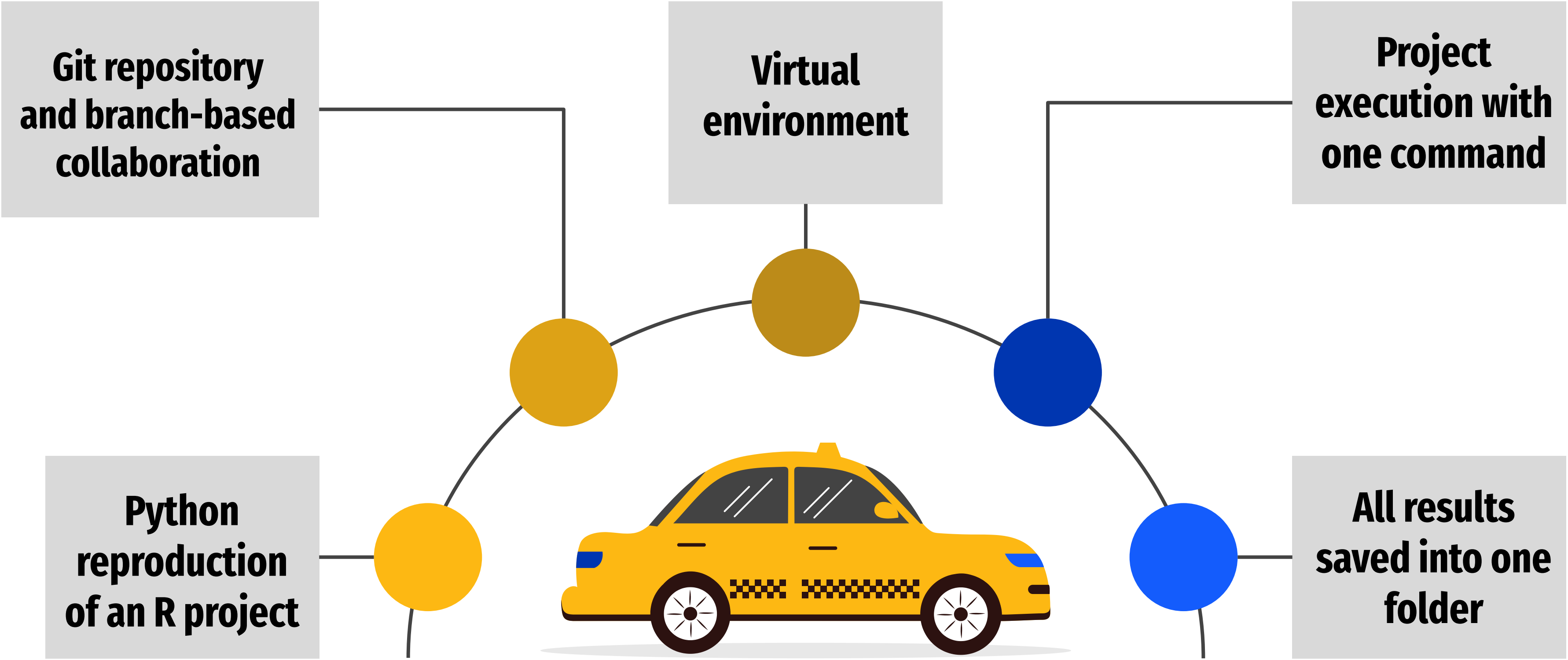
02 Data source - Uber pickups dataset



- Real public Kaggle dataset
- Uber pickup records in New York City/ New Jersey
- Columns: pickup date and time, latitude, longitude, and base information.
- Over 1 million observations

Date/Time	Lat	Lon	Base
09.01.2014 00:01	40.2201	-74.0021	B02512
09.01.2014 00:01	40.75	-74.0027	B02512
09.01.2014 00:03	40.7559	-73.9864	B02512
09.01.2014 00:06	40.745	-73.9889	B02512
09.01.2014 00:11	40.8145	-73.9444	B02512
09.01.2014 00:12	40.6735	-73.9918	B02512
09.01.2014 00:15	40.7471	-73.6472	B02512
09.01.2014 00:16	40.6613	-74.2691	B02512
09.01.2014 00:32	40.3745	-73.9999	B02512

02 Approach


Git repository and branch-based collaboration
Virtual environment
Project execution with one command
Python reproduction of an R project
All results saved into one folder

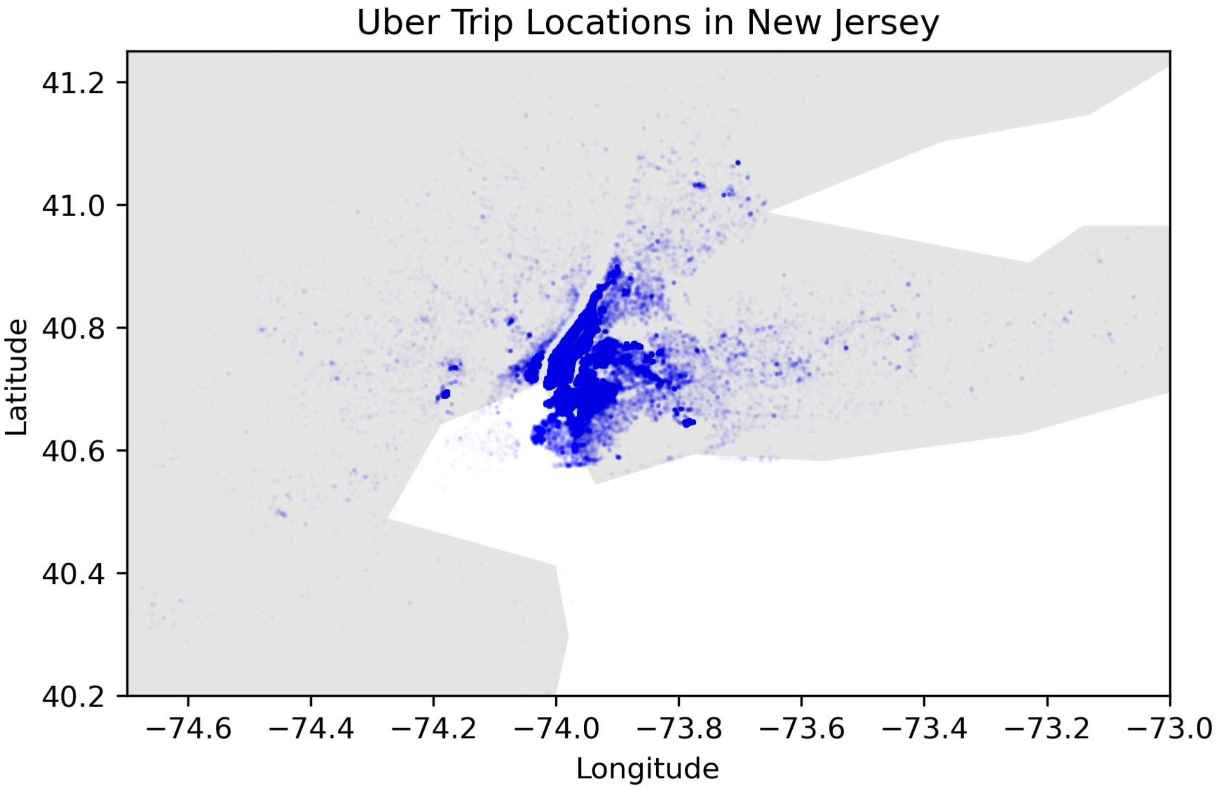
03 Data preparation

Data cleaning and preprocessing, creating additional columns for date and hour

Splitting the dataset into training and testing sets

Data visualisation for explanatory analysis

03 Visualisation of Uber points on the map

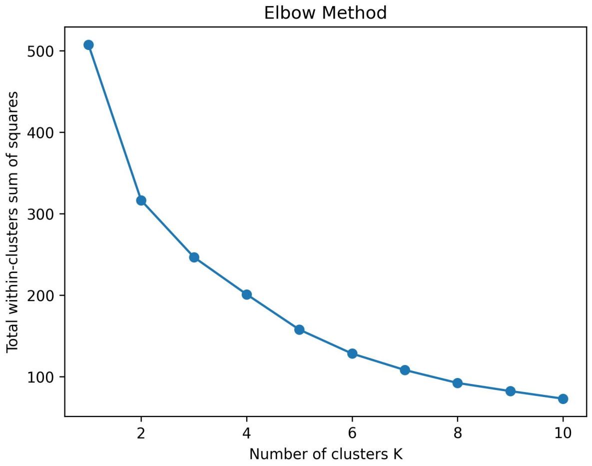


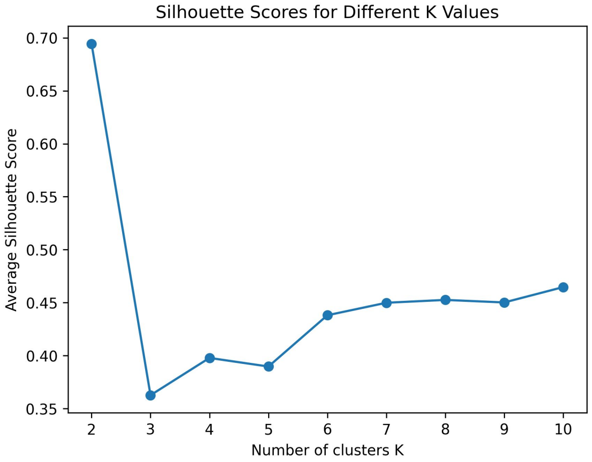
03 Choosing the right number of clusters

01

02

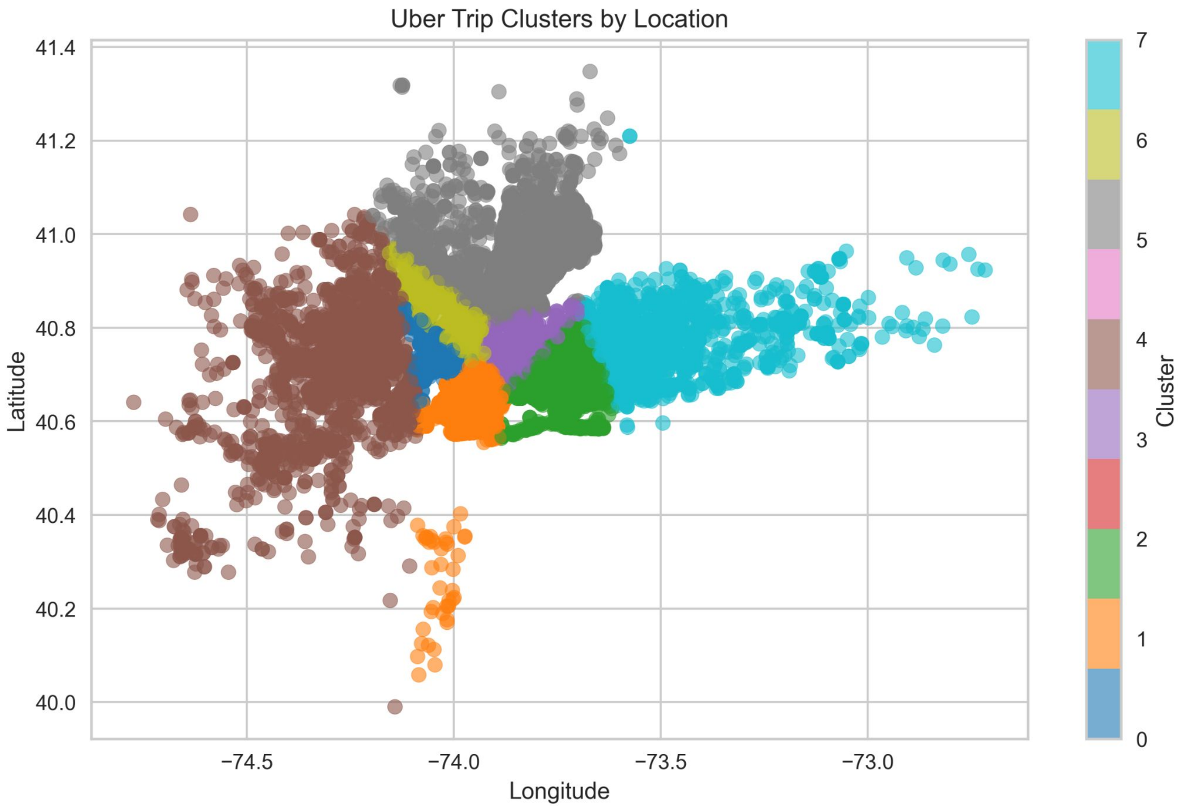
03





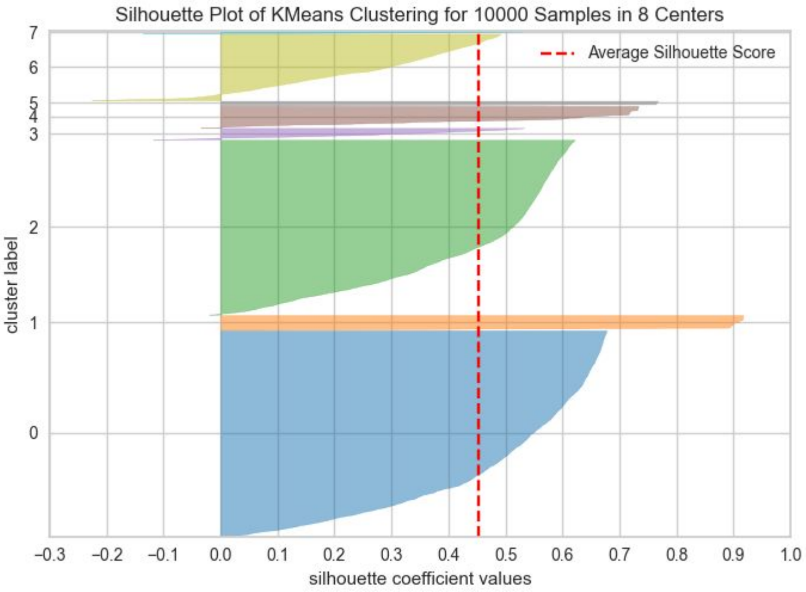
Kmeans Clustering

for the Number of Clusters k = 8

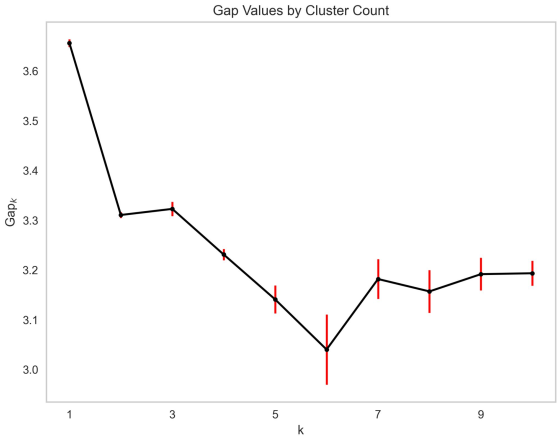


03 Quality measures

01 Silhouette Index



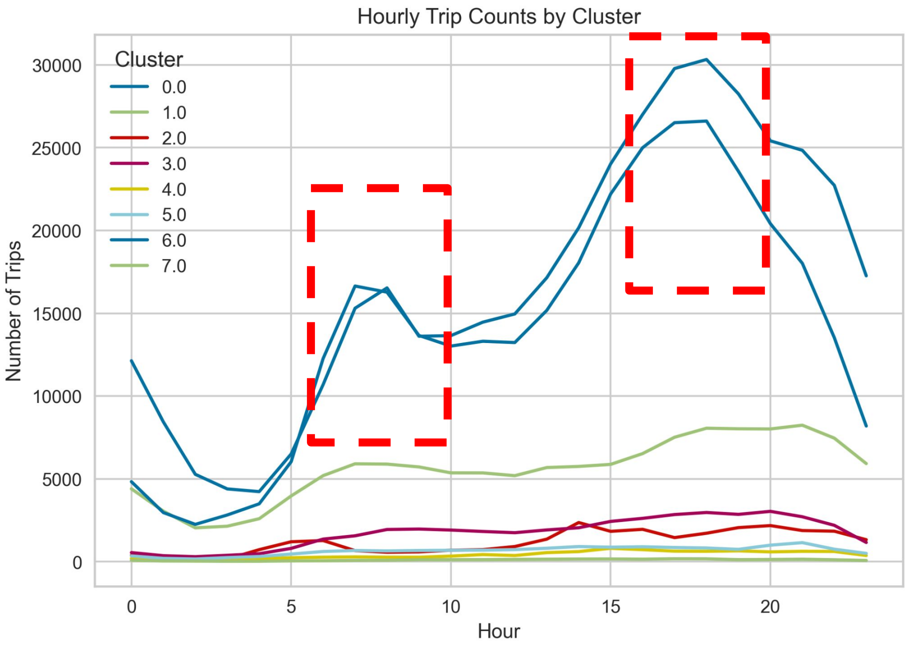
02 GAP Statistic



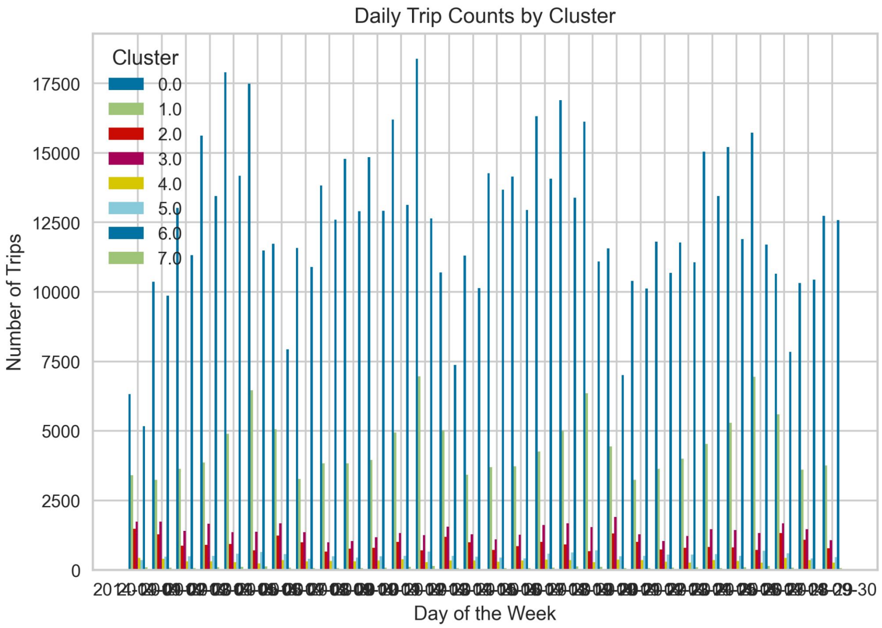
n_clusters	gap_value	ref_dispersion_std	sk	gap*	sk*	diff	diff*
1	3,656	14,021	0,008	1810,405	1819,489	0,352	1814,279
2	3,310	4,951	0,006	773,414	777,287	0,002	776,448
3	3,323	8,884	0,014	594,976	598,010	0,103	597,257
4	3,231	5,269	0,011	451,195	453,476	0,118	453,041
5	3,141	10,145	0,028	339,518	341,363	0,171	341,667
6	3,040	20,721	0,071	272,255	274,404	-0,102	273,608
7	3,182	9,798	0,040	229,242	230,596	0,068	230,464
8	3,157	9,236	0,043	202,576	203,798	-0,002	203,596
9	3,192	6,513	0,033	180,971	181,991	0,023	181,847
10	3,193	4,355	0,025	164,023	164,899		

03 Analysis of Cluster Centers

01 By time

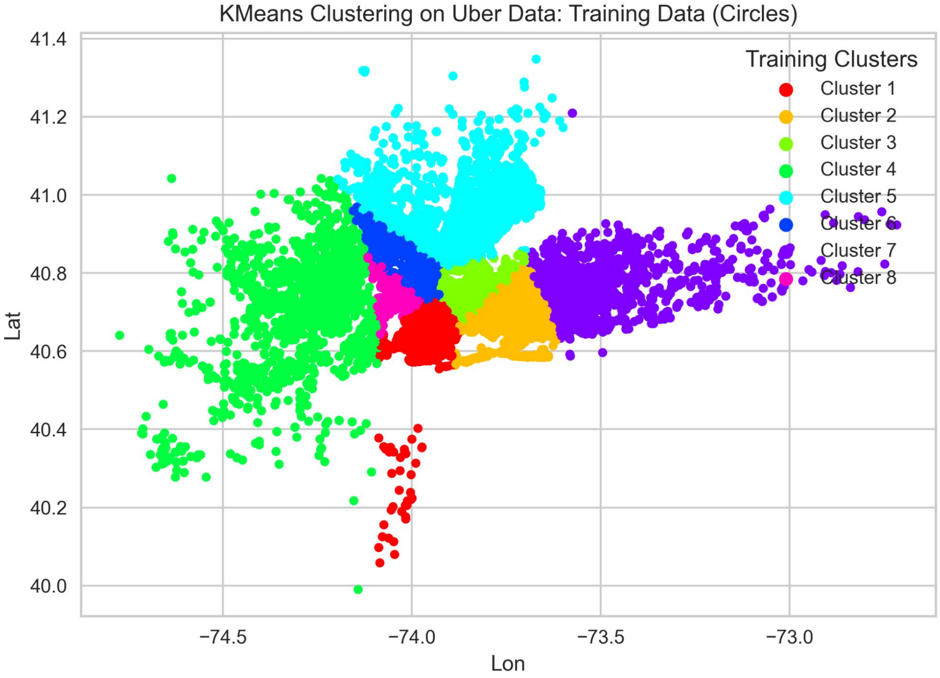


02 By date

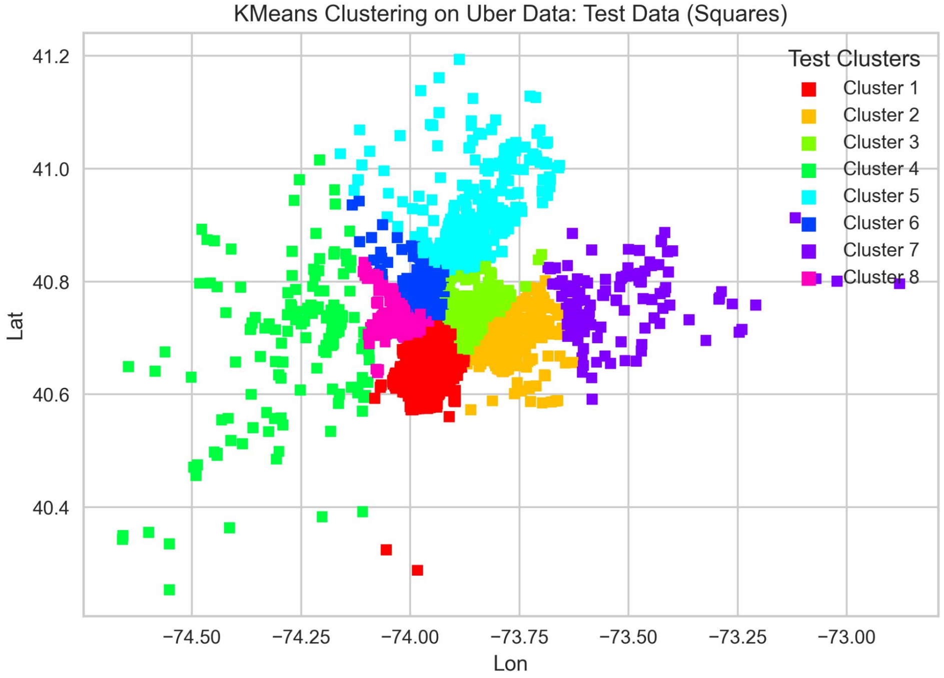


03 Forecasting

01 Kmeans clustering on training data



02 Kmeans clustering on test data



04

Live Demo

Q&A Session